

Autonomous Navigation Prototype for Wheelchair Mobility Assistance Using ROS2 and LiDAR

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Abstract. Mobility is essential for independence and social participation, yet individuals with motor impairments often face challenges navigating large public spaces such as university campuses. In this work, we present the design and implementation of a scaled prototype of a smart wheelchair capable of autonomous navigation. The system is developed using ROS2 Humble and integrates SLAM techniques with Nav2, leveraging a Slamtec RPLiDAR A1 sensor for environment mapping. A differential drive platform simulates the wheelchair’s mobility model, enabling point-to-point navigation based on voice or manual input of campus locations. Preliminary results demonstrate successful navigation in simulation and controlled indoor settings, using maps reconstructed from scanned layouts of the university campus. This work presents a prototype under active development as part of a university social service program to contribute to inclusive mobility solutions aimed at enhancing user accessibility in environments where physical infrastructure is often inadequate and manual operation imposes excessive demands.

Keywords: Autonomous wheelchair, ROS2, SLAM, LiDAR, differential drive, embedded systems.

1 Introduction

Mobility is a fundamental human right and a cornerstone of independence and social inclusion. However, people with limited mobility, whether due to neurological, muscular, or structural immobility, often experience significant barriers to navigation in large public environments such as university campuses. These challenges extend beyond physical difficulty; they impact educational access, social participation, and overall quality of life.

From a technical perspective, traditional assisted mobility devices rely heavily on user control and preinstalled navigational infrastructure, such as floor

markings, or designated signage indicating accessible paths and ramps within buildings.

In contrast, our approach is not dependent on these types of external cues or environmental modifications to operate. Instead, the system relies on onboard sensing and open-source algorithms to autonomously perceive and adapt to its environment. By integrating SLAM (Simultaneous Localization and Mapping) and path planning, the wheelchair can build a map of the environment, localize itself, and determine optimal routes in real time.

The current prototype simulates a differential-drive wheelchair platform equipped with an RPLiDAR A1 sensor, operating within the ROS2 Humble environment. This work bridges technical innovation with social impact, laying the foundation for a navigation system that is not only functionally robust but also deeply aligned with principles of accessibility and autonomy.

One of the key motivators behind this system is the high incidence of wheelchair-related difficulties in university settings, particularly on ramps and sloped access paths. These situations often demand excessive physical effort and can lead to unsafe or stressful experiences for users. By promoting autonomous navigation that does not depend on constant manual input or physical exertion, the system seeks to empower users and mitigate risks associated with underdeveloped accessibility infrastructure.

Although the system is currently in prototype phase, preliminary experiments in real indoor environments have validated core components under constrained conditions.

2 Related Work

2.1 Importance of Prior Work

Developing autonomous mobility solutions for users with physical disabilities requires building on a foundation of prior research in robotics, assistive technology, and human-centered design. Existing literature provides not only technical insights but also identifies critical gaps related to accessibility, scalability, and real-world deployment. This body of work demonstrates the need to move beyond lab-constrained prototypes and toward systems capable of operating in unstructured, dynamic environments—such as university campuses. In this section, we review the state-of-the-art in smart wheelchair systems and the SLAM and navigation frameworks that enable autonomy, while highlighting how our approach addresses unmet needs in this evolving space.

2.2 Smart Wheelchairs: State-of-the-Art

Kapsalis et al. [1] provides an in-depth analysis of urban mobility challenges faced by wheelchair users and emphasizes the decreasing attention to infrastructure efforts that support their mobility needs.

Leaman and La [2] provide a comprehensive review of smart wheelchair developments, classifying systems by levels of autonomy and cost, and emphasizing how sensor integration improves functional mobility for users. Their work categorizes wheelchairs by autonomy level and control strategy, identifying the need for accessible, flexible solutions for unstructured spaces—such as university campuses.

Recent efforts illustrate different approaches to this challenge. For example, the "CoNav Chair" [3] uses shared control and obstacle avoidance within indoor environments to increase user trust and safety. While effective in structured spaces, such systems rely on continuous user input and are limited by environmental assumptions. In contrast, our work aims to enable full autonomy through point-to-point navigation using real-time localization and adaptive path correction mechanisms across outdoor and semi-structured spaces, enabling independent operation without shared control. This shift is critical for scenarios like campus navigation, where dynamic conditions and spatial complexity demand greater autonomy and environmental adaptability.

2.3 SLAM and Navigation Frameworks

Simultaneous Localization and Mapping (SLAM) has become a foundational technique for mobile robots navigating unknown environments. Cadena et al. [4] present a detailed overview of modern SLAM algorithms, discussing issues like data association, long-term operation, and scalability—factors particularly relevant for navigating expansive, partially dynamic environments such as academic campuses. Our system adopts a 2D LiDAR-based SLAM approach to autonomously construct environment maps without prior modifications or external infrastructure.

Caracciolo et al. [5] apply SLAM to a smart wheelchair in indoor settings, eliminating the need for predefined routes or physical guides. Their work influenced our decision to pursue a map-based, infrastructure-light approach, particularly in evaluating the trade-offs between sensor cost and navigational accuracy.

Raspberry Pi-based platforms like "Walk-IT" [6] use SLAM and LiDAR (including RPLiDAR A1) in a modular, cost-effective design. These implementations underscore that a low-cost LiDAR sensor with ROS2 integration is sufficient for accurate obstacle avoidance and mapping—validating the hardware approach of our prototype.

The Robot Operating System 2 (ROS2) is an open-source middleware framework widely used in academic and industrial robotics. Designed to support modular, real-time systems, ROS2 provides tools and libraries for integrating sensors, actuators, navigation algorithms, and control logic in a distributed architecture. Its support for the Nav2 stack and SLAM Toolbox makes it a strong foundation for autonomous mobile systems like smart wheelchairs. — The emergence of ROS2 has further enabled real-time robotic systems to scale effectively. Macenski et al. [7] detail the architecture of the Nav2 stack, which supports modular integration of planning, control, and behavior management

through features like lifecycle nodes and behavior trees. Paired with SLAM Toolbox, which offers robust online and offline SLAM capabilities, this framework provides the computational and algorithmic foundation for our system. These tools make it possible to develop reliable autonomy on constrained platforms, suitable for real-world assistive applications.

3 System Architecture

The proposed system is a scalable, cost-effective prototype of a smart wheelchair capable of autonomous navigation in a semi-structured environment—specifically, a university campus. It integrates key components of modern mobile robotics, including SLAM for environment mapping and localization, global and local path planning via Nav2, LiDAR sensing, and microcontroller-level motor control using micro-ROS. The complete system architecture is designed around the modular and real-time capabilities of ROS2 Humble, and all components are deployed on a NVIDIA Jetson Nano platform.

3.1 Overview of Architecture

The architecture consists of four main subsystems:

- **Sensing and Mapping:** LiDAR + SLAM Toolbox
- **Navigation and Path Planning:** ROS2 Nav2 Stack
- **Low-level Control and Actuation:** Microcontroller + micro-ROS
- **Compute and Integration Platform:** NVIDIA Jetson Nano

Each subsystem is designed to operate semi-independently while exchanging data through ROS2 topics, services, and actions. This ensures robustness and allows real-time decision-making for both global and local tasks.

3.2 Environment Mapping and Localization

We implemented SLAM Toolbox, a ROS2-compatible library, to autonomously generate an occupancy grid map of the environment. Initial tests used scanned floorplans of the university campus, provided by the VANTEC student group, as visual references for simulation and map alignment. These maps were not directly integrated into SLAM but served to guide environment modeling.

The RPLiDAR A1 from Slamtec was chosen for its affordability and compatibility with ROS2. Operating at 5–10 Hz with 360° coverage, the LiDAR sensor enables the system to:

- Continuously correct localization during motion,
- Update the internal map as new features appear (e.g., temporary obstacles),
- Enable recovery from planning errors via loop closure and local corrections.

This dual approach of static map reference (for global planning) and dynamic SLAM (for real-time adaptation) enables reliable autonomy in dynamic spaces such as corridors, walkways, and open campus areas.

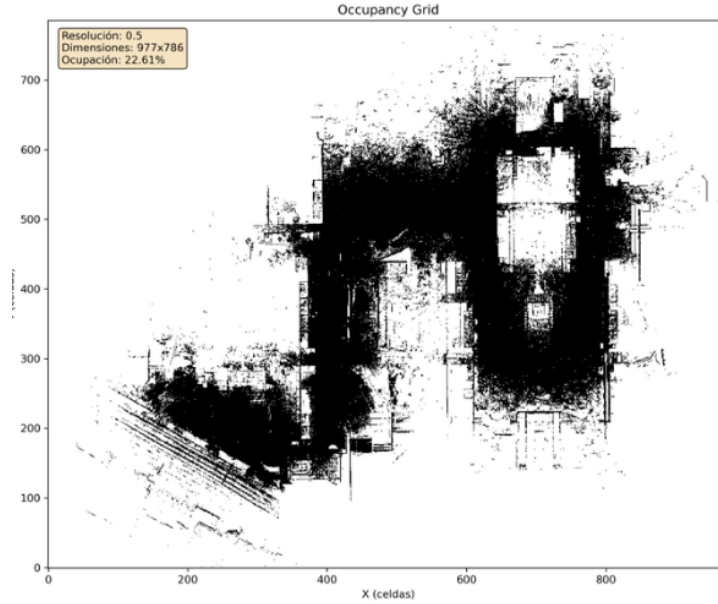


Fig. 1. Occupancy grid map aligned with scanned campus layout.

3.3 Global and Local Navigation

Autonomous movement is enabled through Nav2, the official ROS2 navigation stack. The system is structured around two planning layers:

- **Global Planner:** Computes an optimal path from the current location to the target pose using the static map. This layer uses A* or Smac Planner algorithms over the occupancy grid.
- **Local Planner:** Continuously refines and adjusts the global path using real-time LiDAR data. The DWB (Dynamic Window Approach) controller evaluates possible trajectories to avoid dynamic obstacles like pedestrians, bicycles, and animals.

This hierarchical planning model aligns with best practices in mobile robotics, ensuring the robot can both:

- Efficiently reach a destination based on precomputed routes,
- React to unexpected environmental changes by adjusting its path on-the-fly.

Such adaptability is critical for campus deployment, where environments are inherently unpredictable.

3.4 Onboard Computation

All high-level processing is handled by an NVIDIA Jetson Nano, selected for its low power consumption and ability to run ROS2 nodes, LiDAR drivers, and SLAM algorithms in real time. The Jetson executes:

- SLAM Toolbox for map construction and localization,
- Nav2 stack for path planning and behavior tree execution,
- Sensor fusion nodes and transform broadcasting (TF2).

The Jetson also acts as the bridge between high-level navigation logic and low-level motor control via serial communication with the microcontroller.

3.5 Motor Control and Embedded Integration

To actuate the differential-drive system, we employed a microcontroller running micro-ROS with RTOS. This setup enables:

- ROS2-compatible messaging over serial,
- Real-time execution of velocity commands (`cmd_vel`) from Nav2,
- Encoder feedback processing for odometry estimation.

The differential drive configuration simulates real-world wheelchair actuation, where the left and right wheels are controlled independently to enable both linear and angular motion. This model is particularly well-suited for confined or cluttered environments like ramps, narrow hallways, and student pathways.

3.6 System Integration and Flow

The complete data flow is summarized as follows:

1. **User Input:** The user selects or provides a goal location (via command-line or GUI interface).
2. **Global Planning:** Nav2 computes a path to the destination using the SLAM-generated static map.
3. **LiDAR Sensing:** The RPLiDAR continuously scans the environment, feeding data into SLAM Toolbox and the local planner.
4. **Localization and Corrections:** SLAM refines the robot's pose; Nav2 replans if significant deviation occurs.
5. **Control Execution:** Velocity commands are sent to the microcontroller via micro-ROS.
6. **Motion:** The motors execute the commands and encoder data updates odometry for closed-loop control.

This flow ensures autonomy with high reactivity, achieving reliable navigation in real-world, unpredictable conditions.

4 Implementation Details

The implementation of our autonomous wheelchair prototype was developed within the ROS 2 Humble framework and deployed on an NVIDIA Jetson Nano platform running Ubuntu 22.04. The system integrates official ROS 2 packages alongside custom-developed nodes to perform sensing, localization, path planning, and low-level control. This section provides an overview of the primary components and describes how they interact within the system.

4.1 LiDAR Integration and SLAM

To provide environmental perception and real-time map updates, we utilized the RPLiDAR A1 2D laser scanner by Slamtec. The ROS 2 driver `rplidar_ros2`, officially provided and maintained by Slamtec, was used to interface with the sensor. This node publishes laser scan data (`/scan`) and broadcasts necessary transforms between the LiDAR frame and the robot base frame.

These laser scans are processed by SLAM Toolbox, configured in asynchronous mode for online mapping and localization. The resulting occupancy grid map (`/map`) and estimated pose (`/slam_toolbox/pose`) are published continuously and used both for global path planning and local corrections. This allows the system to detect and respond to unexpected dynamic obstacles and minor trajectory deviations in real time.

4.2 Navigation Stack

The ROS 2 Nav2 stack was configured to enable autonomous movement toward user-defined goal locations. The main nodes used in this setup include:

- `nav2_bt_navigator`: Orchestrates behavior tree execution and goal management.
- `nav2_controller`: Implements local control using the DWB (Dynamic Window Approach) plugin.
- `nav2_planner`: Computes global paths over the static or SLAM-generated map.

The navigation pipeline combines global path planning based on the static occupancy grid with local real-time corrections from LiDAR scans. This separation of global and local layers aligns with common robotics paradigms, allowing the robot to avoid emergent obstacles such as people, animals, or incorrectly planned routes (e.g., blocked corridors or temporary construction).

4.3 Localization and Odometry Tracking

In addition to SLAM-based localization, we implemented a custom ROS 2 node called `wheelchair_odometry`. This node processes encoder feedback from the differential-drive motors to compute odometry data, which is published to the `/odom` topic. The node is responsible for:

- Estimating the robot’s relative position and velocity,
- Maintaining accurate tracking when LiDAR data is momentarily degraded or occluded,
- Feeding odometry data into the transform tree (`/tf`) for integration with Nav2 and SLAM Toolbox.

This redundancy between odometry and SLAM enhances the system’s reliability and ensures smoother navigation, particularly in cases of brief sensor occlusion or during re-localization events.

4.4 Embedded Control and Motor Actuation

The low-level actuation is managed by a microcontroller running micro-ROS, which allows ROS 2 communication over a serial interface. The controller receives velocity commands (`/cmd_vel`) from the navigation stack and translates them into PWM signals to actuate the left and right motors of the differential drive system.

The differential-drive model was selected to closely replicate real-world wheelchair motion, where independent wheel control allows precise turning and maneuvering in confined environments. Encoder feedback is read by the microcontroller and transmitted back to the Jetson for odometry estimation, closing the control loop.

4.5 Development Status and Considerations

As of this writing, the prototype is in an advanced development stage, with all major architectural components integrated and functional under experimental conditions. The system has been tested successfully in simulated and partially mapped real environments; however, several elements remain in refinement:

- Full physical deployment across the complete campus map is in progress,
- User interaction interfaces (e.g., touchscreen or voice input) are currently being developed,
- Additional failsafe behaviors and recovery strategies are under review for unpredictable scenarios.

Rather than representing a final implementation, this work illustrates a robust, modular foundation for an assistive autonomous mobility system. Ongoing development will focus on expanding real-world testing, improving reliability, and optimizing resource usage on embedded hardware.

5 Evaluation

At the current stage of development, our autonomous wheelchair system remains in prototype form, with most evaluations performed in controlled environments and simulation. These tests aim to verify the performance of core subsystems: SLAM-based mapping, odometry tracking, local and global planning, and motor actuation via micro-ROS. A limited number of real-world experiments have also been conducted in simplified indoor settings to validate key functionalities under physical constraints.

5.1 Simulation-Based Evaluation

Using Gazebo and RViz within the ROS 2 ecosystem, we conducted extensive simulation trials to test:

- SLAM Toolbox’s ability to construct and update occupancy grids from LiDAR input,
- Nav2’s global planning across a semi-structured 2D map (based on a scanned section of the campus),
- The system’s responsiveness to simulated dynamic obstacles, such as moving pedestrians or obstructions.

In these trials, the prototype successfully generated consistent maps and followed global paths with minor local corrections. Obstacle avoidance using the DWB local planner performed reliably when navigating around dynamic elements, validating the effectiveness of our layered navigation approach (global + local).

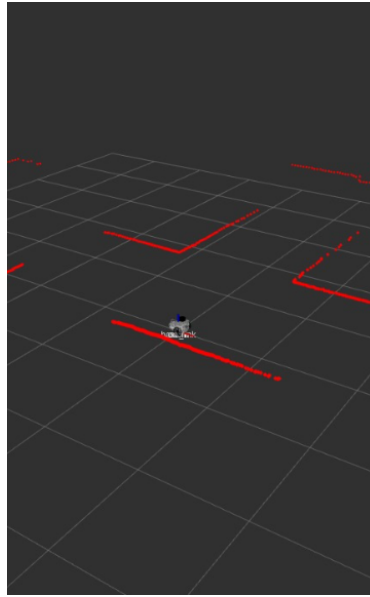


Fig. 2. SLAM and navigation visualization during simulation using Gazebo and RViz.

5.2 Physical Testing

Initial physical evaluations took place in a controlled 2.4×2.4 meter indoor environment resembling classroom corridors. These tests aimed to:

- Verify that encoder-based odometry (via `wheelchair_odometry`) remained synchronized with SLAM estimates,

- Validate communication and timing between ROS 2 nodes running on the Jetson Nano and the microcontroller via micro-ROS,
- Observe behavior when encountering physical, unmodeled obstacles such as chairs or people.

Results indicate that:

- SLAM and odometry are sufficiently accurate for short- to medium-range navigation in known areas,
- The system reacts appropriately to unexpected obstacles by re-planning or stopping,
- Communication between nodes is stable, although system latency increases slightly under higher processing load.

These are best described as preliminary results, demonstrating functional integration of core modules; however, full-scale deployment and long-term performance remain under evaluation. Nevertheless, they confirm the viability of the proposed architecture and justify further refinement.

5.3 Limitations and Observations

Several important limitations have been identified:

- Path planning at larger campus scales still requires refinement, especially in handling long-term dynamic changes (e.g., construction zones or high foot traffic),
- Real-time performance under moderate computational load remains acceptable, but becomes constrained under high-load conditions—especially during concurrent execution of SLAM and path planning,
- Point-to-point navigation has been validated in short-range use cases, but user-facing goal-setting and persistent multi-session maps are still under development.

These findings suggest that improving the onboard hardware could significantly enhance system responsiveness and enable support for more complex environments and behaviors.

6 Conclusions and Future Work

This paper presents an assistive robotic prototype for autonomous wheelchair navigation using ROS 2, SLAM, and real-time path planning. Designed as part of a university social service project, the system integrates low-cost hardware, open-source frameworks, and differential-drive mobility to address the needs of individuals with limited mobility in campus-like environments.

Preliminary results—derived from simulation and early physical testing—demonstrate the feasibility of autonomous SLAM-based navigation and obstacle avoidance

using layered planning strategies. The use of ROS 2 components such as SLAM Toolbox, Nav2, has enabled a modular architecture that can be deployed on embedded systems with real-time performance.

Future work will focus on several key areas:

- Scaling the system to a full-size wheelchair capable of fully autonomous point-to-point navigation,
- Expanding real-world deployment across the entire university campus, including map generalization and persistent localization,
- Enhancing system responsiveness by upgrading the computational hardware,
- Developing user-friendly goal-setting interfaces such as voice commands or touchscreen input,
- Improving recovery and fail-safe strategies for scenarios involving localization loss or prolonged obstacle blockage.

The proposed system represents a promising step toward scalable and accessible autonomous mobility solutions.

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